Name: Vallari Ravikiran Panvalkar
Roll No: 51
Experiment No.5
Design and Development of a Web-Based Quiz System Using JavaScript
Date of Performance: 11/02/2026
Date of Submission: 11/20/2026

Vidyavardhini's College of Engineering & Technology
Department of Computer Engineering

Aim:- To design and develop a web-based quiz system using JavaScript.

Objective: To create an interactive quiz application using JavaScript that includes question display, user input handling, score calculation, and dynamic result generation.

Theory:

A web-based quiz application is an interactive platform that allows users to answer questions and receive instant feedback. JavaScript plays a key role in building such applications by dynamically displaying questions, capturing user input, and controlling the flow of the quiz. It enables real-time interaction without requiring page reloads, making the experience smooth and engaging.

The application includes features such as presenting questions one by one, validating user answers, calculating scores, and generating results dynamically. JavaScript functions and event handling are used to manage user responses and update the interface accordingly. This helps in creating an efficient and user-friendly quiz system that provides immediate results and enhances learning or assessment experiences.

Implementation:

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Experiment 5 - Quiz System</title>

  <style>

    body { margin: 0; font-family: Arial, sans-serif; background: #f1f5f9; }
```

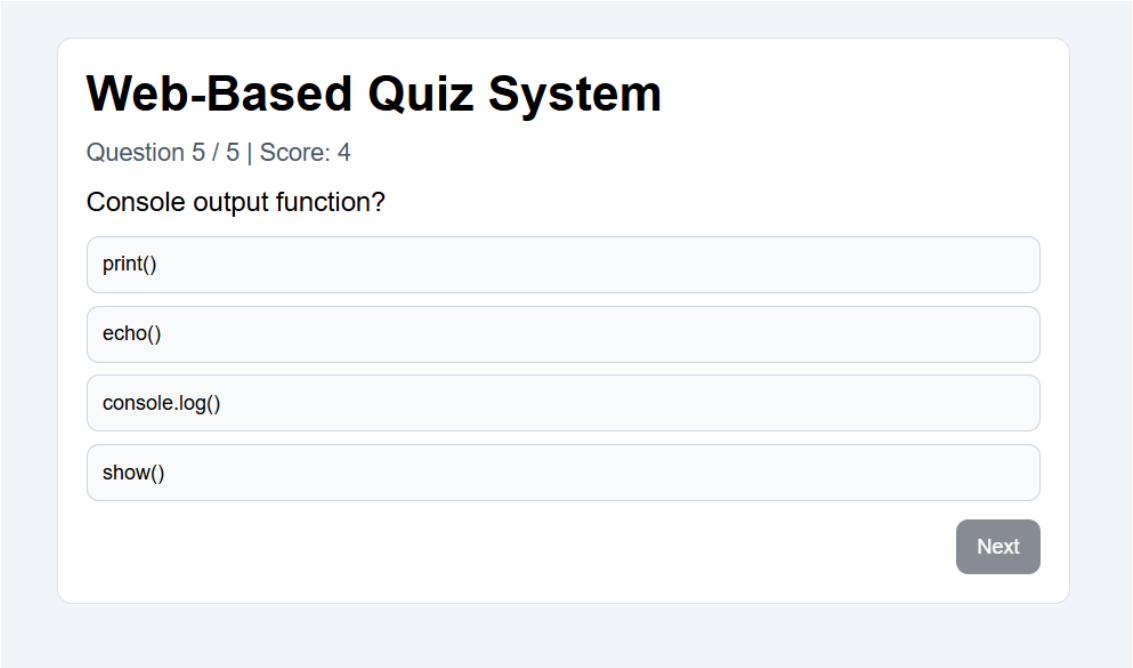
```
    .box { width: min(620px, 94vw); margin: 28px auto; background: #fff; border: 1px solid
#dbe2ea; border-radius: 10px; padding: 18px; }

    h1 { margin: 0 0 10px; }

    #meta { color: #475569; margin-bottom: 12px; }

    #q { font-size: 1.1rem; margin: 0 0 12px; }

    #opts { display: grid; gap: 8px; }

    #opts button { border: 1px solid #cbd5e1; background: #f8fafc; border-radius: 8px;
padding: 10px; text-align: left; cursor: pointer; }

    #opts button.correct { background: #dcfce7; border-color: #22c55e; }

    #opts button.wrong { background: #fee2e2; border-color: #ef4444; }

    .row { margin-top: 12px; display: flex; justify-content: space-between; align-items:
center; gap: 10px; flex-wrap: wrap; }

    #next, #restart { border: 0; background: #0f172a; color: #fff; border-radius: 8px; padding:
10px 14px; cursor: pointer; }

    #next:disabled { opacity: 0.5; cursor: not-allowed; }

    .hide { display: none; }

  </style>

</head>

<body>

  <div class="box">

    <h1>Web-Based Quiz System</h1>

    <div id="meta">Question 1 / 5 | Score: 0</div>

    <p id="q"></p>

    <div id="opts"></div>

    <div class="row">

      <span id="msg"></span>

      <button id="next" disabled>Next</button>

      <button id="restart" class="hide">Restart</button>
```

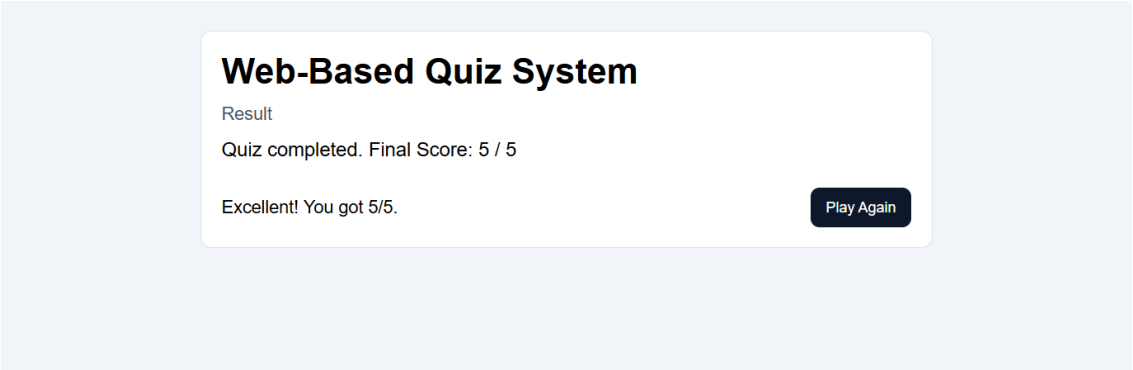
```
    </div>

  </div>



  <script>

    const quiz = [

      { q: "Which tag includes JavaScript?", o: ["<css>", "<script>", "<js>", "<link>"], a: 1 },

      { q: "DOM stands for?", o: ["Document Object Model", "Data Object Method",
"Display Oriented Module", "Document Output Map"], a: 0 },

      { q: "Which one is strict equality?", o: ["==", "=", "===", "!="], a: 2 },

      { q: "Method to select element by id?", o: ["getById()", "getElementById()", "query()",
"selectId()"], a: 1 },

      { q: "Console output function?", o: ["print()", "echo()", "console.log()", "show()"], a: 2
}

    ];



    const meta = document.getElementById('meta');

    const qEl = document.getElementById('q');

    const opts = document.getElementById('opts');

    const msg = document.getElementById('msg');

    const next = document.getElementById('next');

    const restart = document.getElementById('restart');



    let i = 0, score = 0, locked = false;



    function render() {

      locked = false; next.disabled = true; msg.textContent = '';

      meta.textContent = `Question ${i + 1} / ${quiz.length} | Score: ${score}`;
```

```
  qEl.textContent = quiz[i].q;
  opts.innerHTML = '';
  quiz[i].o.forEach((text, idx) => {
    const b = document.createElement('button');
    b.textContent = text;
    b.onclick = () => pick(idx, b);
    opts.appendChild(b);
  });
}

function pick(idx, btn) {
  if (locked) return;
  locked = true;
  [...opts.children].forEach((b, bi) => {
    b.disabled = true;
    if (bi === quiz[i].a) b.classList.add('correct');
  });
  if (idx === quiz[i].a) { score++; msg.textContent = 'Correct'; }
  else { btn.classList.add('wrong'); msg.textContent = 'Wrong'; }
  meta.textContent = `Question ${i + 1} / ${quiz.length} | Score: ${score}`;
  next.disabled = false;
}

next.onclick = () => {
  i++;
  if (i < quiz.length) render();
```

```
      else {
        qEl.textContent = `Quiz completed. Final Score: ${score} / ${quiz.length}`;
        opts.innerHTML = '';
        if (score === quiz.length) {
          msg.textContent = 'Excellent! You got 5/5.';
          restart.textContent = 'Play Again';
        } else {
          msg.textContent = 'Try Again';
          restart.textContent = 'Try Again';
        }
        next.classList.add('hide');
        restart.classList.remove('hide');
        meta.textContent = 'Result';
      }
    };

    restart.onclick = () => {
      i = 0; score = 0;
      next.classList.remove('hide');
      restart.classList.add('hide');
      render();
    };
    render();
  </script>
</body>
</html>
```

Output:





Conclusion:

An interactive quiz application using **JavaScript** displays questions and options dynamically on the webpage. It handles **user input** by recording selected answers and compares them with correct ones to calculate the **score**. JavaScript also manages navigation between questions and generates the **final result dynamically**, making the quiz responsive and user-friendly.